



New developments

February 5-6, 2024

Outline

Here we'll discuss new developments at FloraPulse!

- Small sensor for small crops
 - Installation
 - Test results
 - Upcoming trials
- Heading cut install for avocado, walnut, pecan
- Potential venues: soil microtensiometer, osmometer
- SWP closed-loop irrigation shown possible, and beneficial!

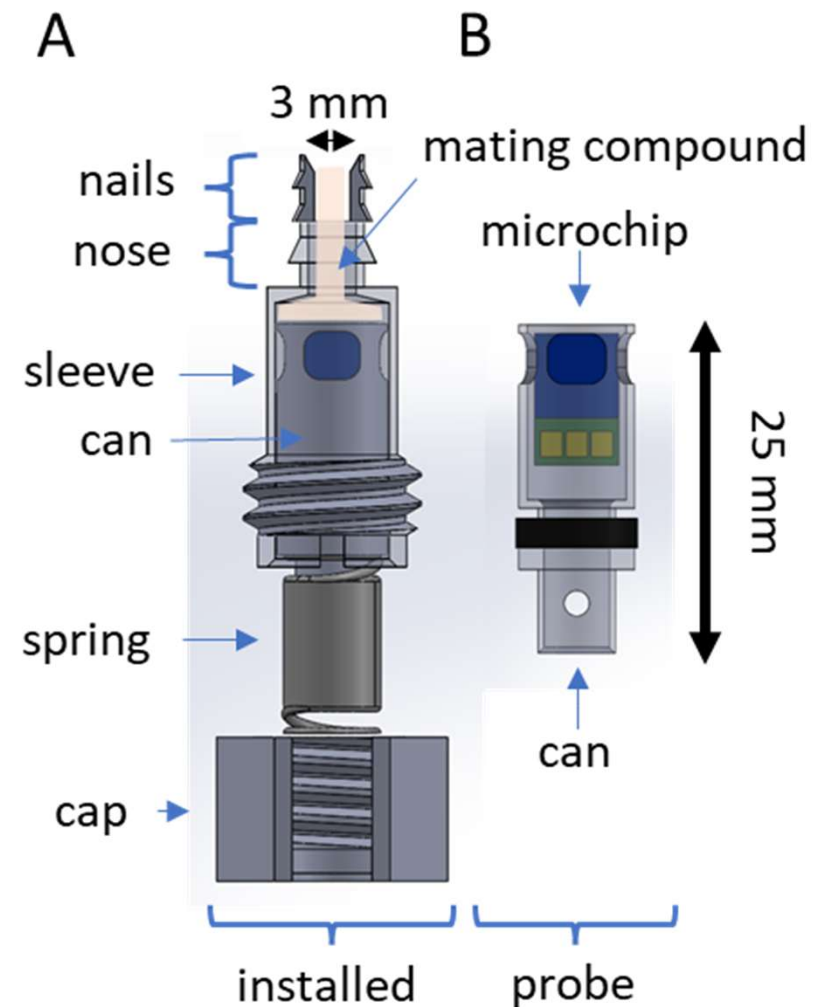
Small sensor

- New smaller sensor and installation for use in stems down to ~1.2 cm in diameter.
- Still under development, but we're working with testers. Interested? Let's discuss.
- Improved:
 - Installation into small stems, more crops accessible.
 - Probably works better for large trees too.
 - Designed for reusability.
 - Smaller wound may increase reliability and accuracy.
 - Use of sealant will create better seal, especially during rain events.



Small sensor – how it works

- Everything shrunk down compared to normal sensor. Sleeve pushed into xylem, holds sensor inside. Fluid exchange through the kaolin clay compound.



Small sensor installation



Pick trunk



Attach guide



Place sleeve on piston

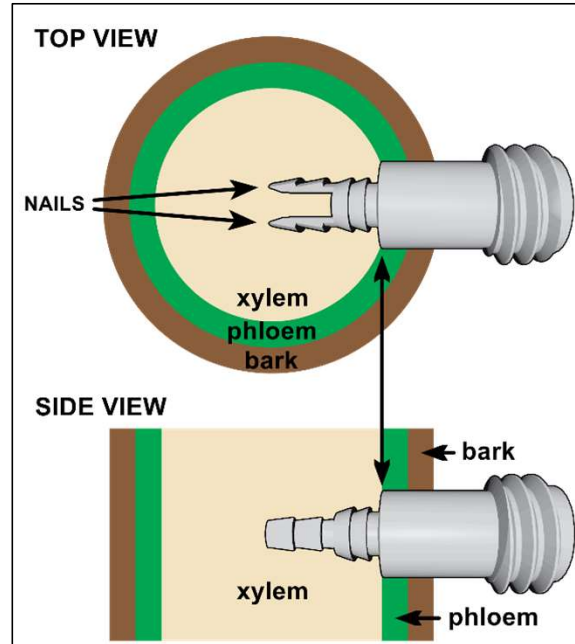


Place piston inside guide and align

Installation #2



Press into place with C-clamp



Installation #3 – insert probe and cover



Insert probe and follow with cap.

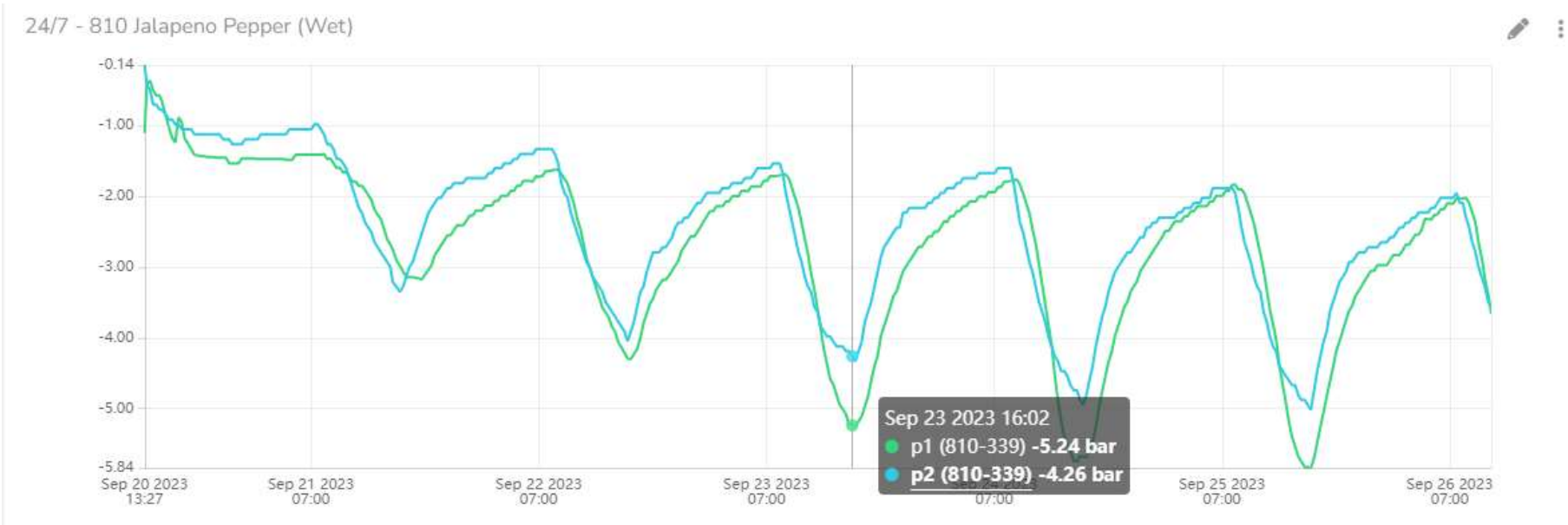


Add sealant
(not grease, photo outdated)

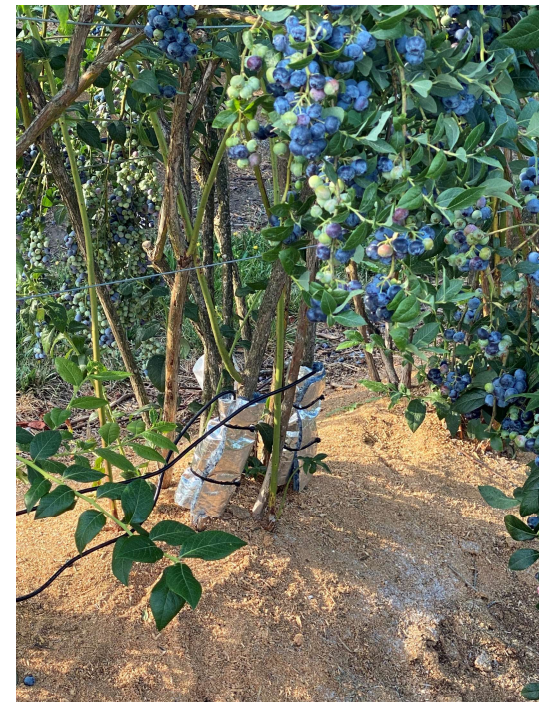
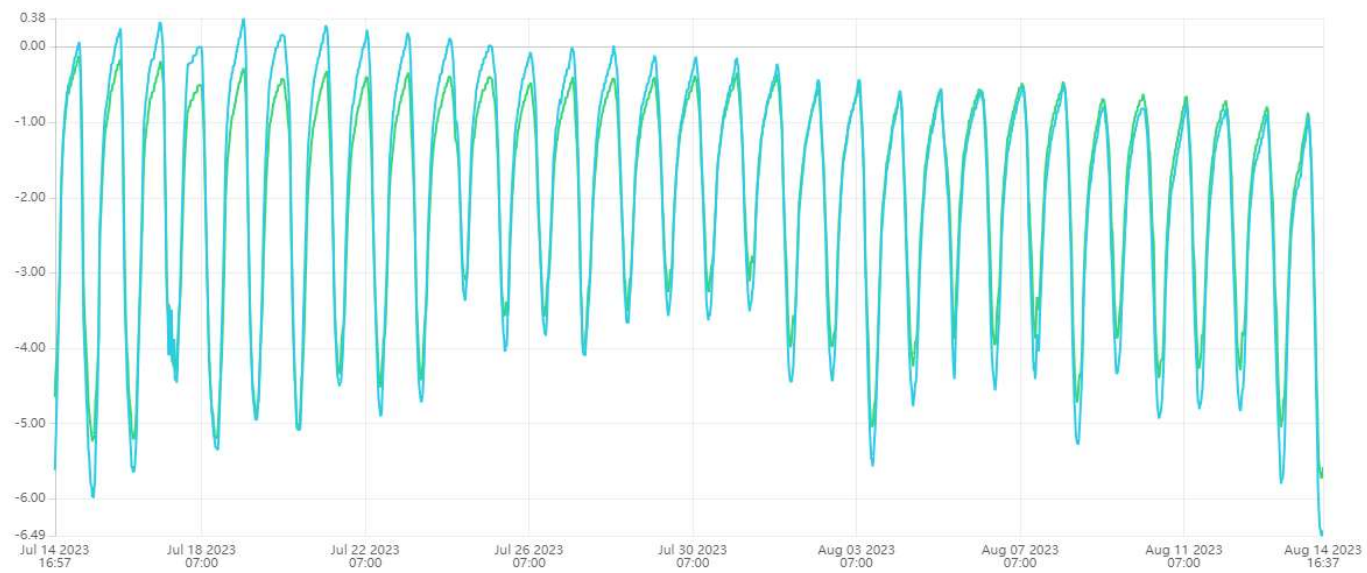


Add insulation

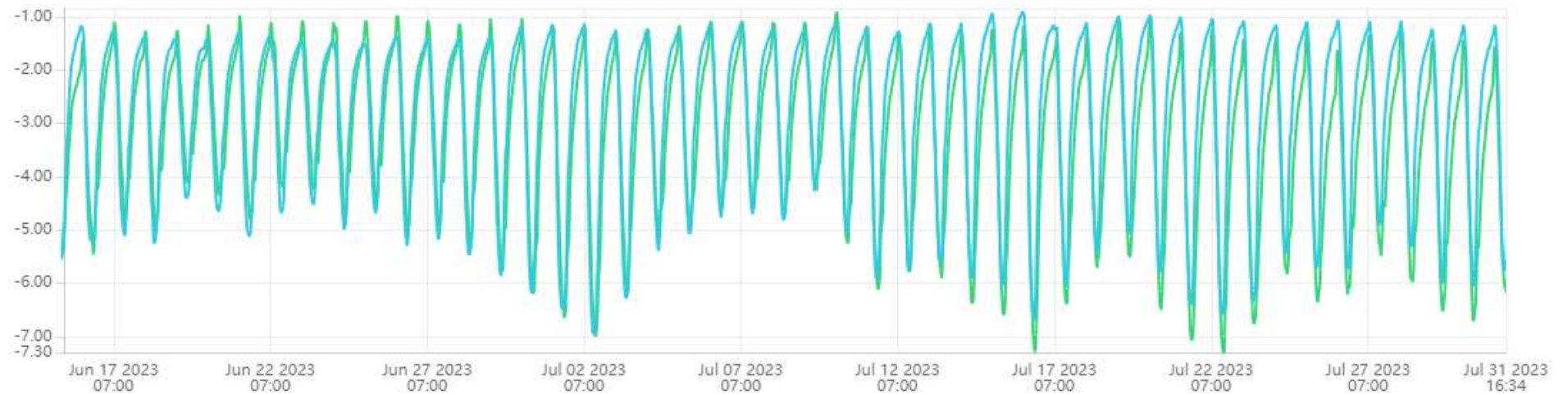
Bell Pepper



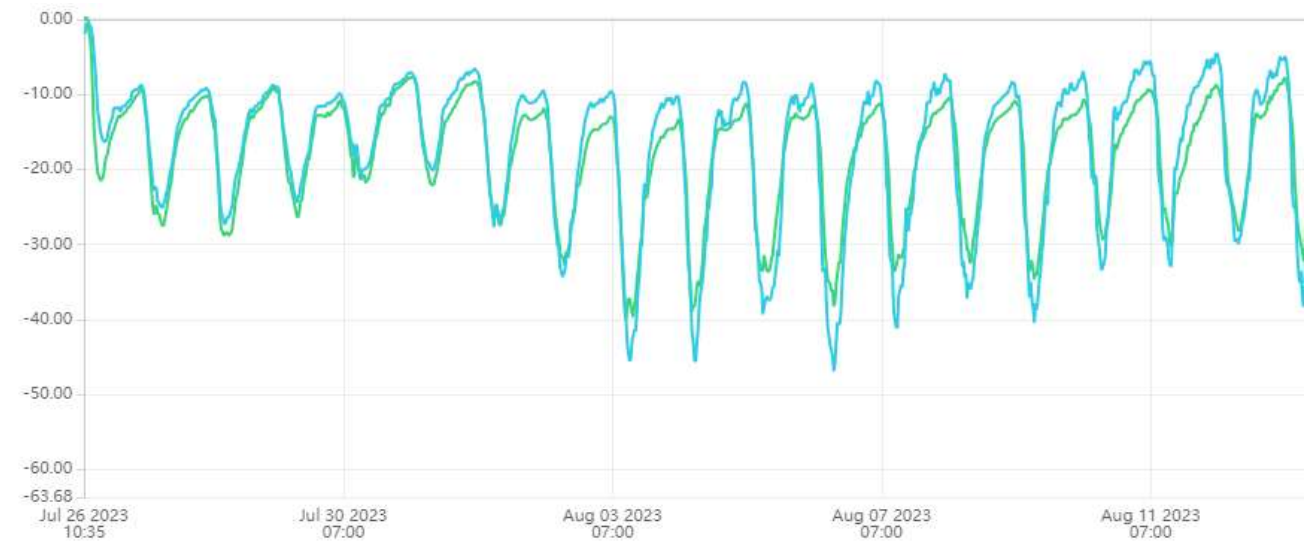
Blueberry



Grape



Cotton



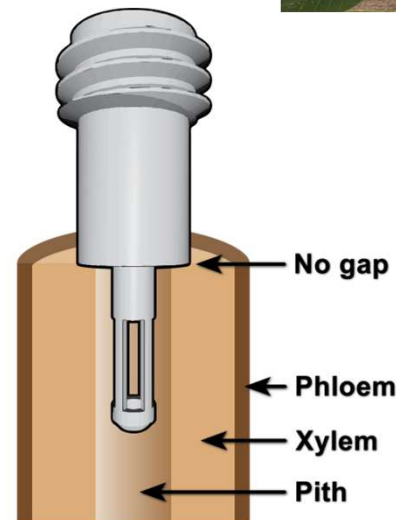
We're doing sales for limited initial users of the small probe this summer.

Contact your FloraPulse distributor!

Hands on: install small sensor

Heading cut for walnut, avocado and pecan

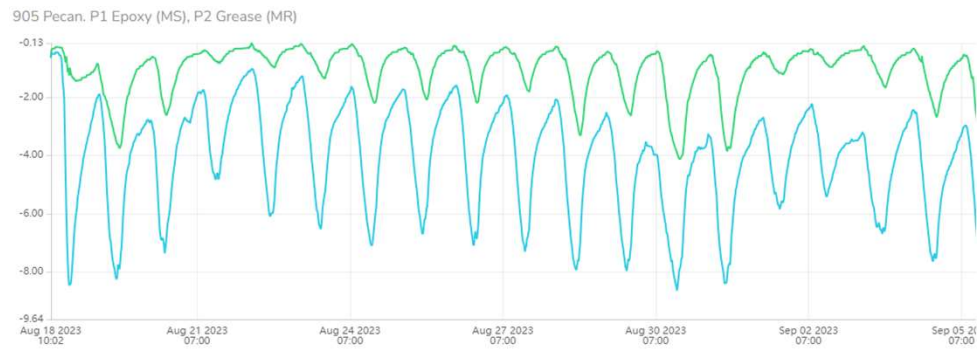
- Walnut, avocado, pecan show very strong wounding response to sensor installation. The sensor usually floods within a week or two and reads zero.
- In collaboration with UC Davis, we developed a new installation style that might avoid this issue.
- Sensor installed in the branch pith, to avoid excessive wounding response.
- Testing so far is promising, but there are still challenges. We expect 2024 trials to be conclusive.



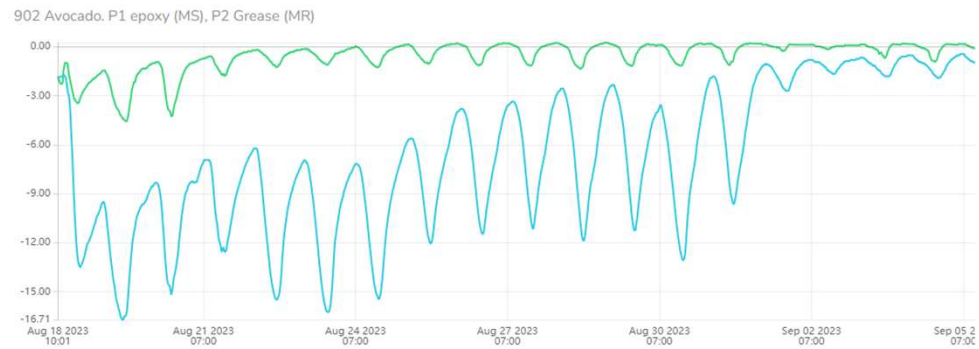
Heading cut representative data



Walnut



Pecan



Avocado

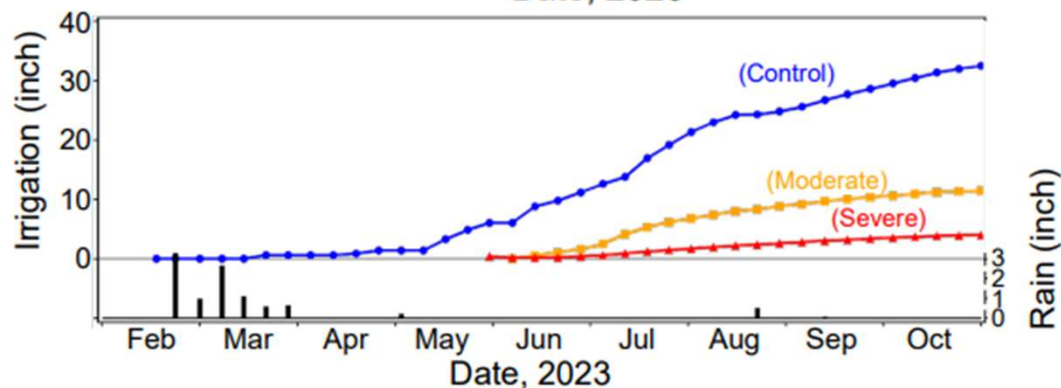
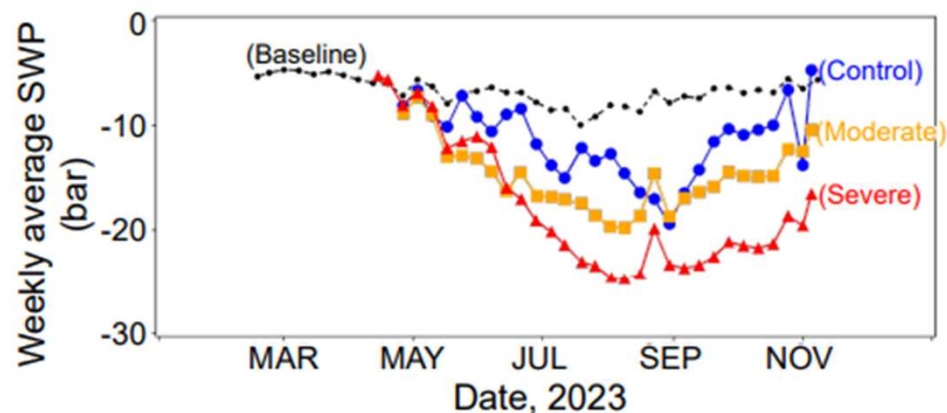
Hands on: show heading cut install

Closed-loop irrigation: promising results

- Professor Ken Shackel from UC Davis set up a closed-loop irrigation system based on FloraPulse SWP data in almonds.
- Lots of industry interest, but need development partner to build into commercial product.

Automatically irrigating to a target level of stress resulted in a very large reduction in total seasonal irrigation.

- Control: 32.5" irrigation
- Moderate stress: 11.5" irrigation
- Severe stress: 4" irrigation



Would you be interested in:

- A microtensiometer soil sensor with
 - Small size
 - Large measurement range (-35 bar).
- A dip-in osmometer.

These are products FloraPulse could develop with the existing technology base, if there is enough interest.